Omar Abu-Ella

Professor of Electrical and Electronic Engineering
Wireless Communications | Massive MIMO | Reconfigurable Intelligent Surfaces | 6G Networks
Misurata, Libya | Phone: +218 91 977 0898 | Email: omarabuella@eng.misuratau.edu.ly | omarabuella@gmail.com

Professional Profile

Professor of Electrical and Electronic Engineering with more than two decades of academic, research, and leadership experience in wireless and mobile communications. Research contributions span interference mitigation and group decoding, multi-antenna and massive MIMO systems, adaptive beamforming, reconfigurable intelligent surfaces (RIS), millimeter-wave communications, NOMA, cell-free architectures, UAV and low-altitude platform communications, and machine-learning-driven optimization of next-generation (6G) radio networks. Doctoral training at the University of Vaasa (Finland) with advanced research residency at Columbia University (USA). Extensive record of peer-reviewed publication, graduate supervision, curriculum development, editorial and technical program committee service, and senior academic administration, including service as Vice Dean for Academic Affairs and Head of Department.

Core Research Areas

6G and intelligent wireless networks	Reconfigurable intelligent surfaces (RIS/IRS), integrated sensing and communication (ISAC), multi-agent reinforcement learning for network control, cell-free massive MIMO, non-terrestrial networks and LEO satellite systems.
Wireless theory and signal processing	Interference mitigation, group decoding, capacity and achievable-rate analysis, channel and path-loss modeling, adaptive antennas, beamforming, hardware-impairment analysis.
Radio access technologies	Massive/XL-MIMO, millimeter-wave and sub-THz communications, NOMA, CoMP, software-defined radio, UAV-assisted and low-altitude platform communications.

Education

Doctor of Science in Technology (D.Sc. Tech.), Telecommunication Engineering 2014
University of Vaasa, Vaasa, Finland
Dissertation: *Interference Mitigation Using Group Decoding in Multiantenna Systems*. Adviser: Prof. Mohammed Elmusrati. Opponent: Prof. Markku Juntti (University of Oulu). Pre-examiners: Prof. Riku Jäntti (Aalto University) and Prof. Majdi Ashibani (Libyan Academy for Telecom and Informatics).

Professional Degree, Electrical Engineering Jan. 2010 – May 2012
Columbia University in the City of New York, New York, NY, USA
Advanced research residency. Research topic: *Group Decoding*. Adviser: Prof. Xiaodong Wang.

M.Sc., Electronics and Electrical Engineering Apr. 2004 – Aug. 2007
Higher Institute of Industry, Misurata, Libya
Thesis: *Hybrid Adaptive Beamforming Techniques for Mobile Cellular Systems*. Adviser: Prof. Bashir El-Jabu.

B.Sc., Electronics and Electrical Engineering Nov. 1997 – Sept. 2001
Academy of Aero Studies and Science, Misurata, Libya
Thesis: *Horn and Parabolic Antenna Design*. Adviser: Prof. Bashir El-Jabu.

Academic Appointments

Professor Dec. 2025 – present
Department of Electrical and Electronic Engineering, Misurata University, Libya
Full professorial rank with responsibilities in research, postgraduate supervision, teaching, and institutional service.

Associate Professor Dec. 2021 – Nov. 2025
Department of Electrical and Electronic Engineering, Misurata University, Libya

Assistant Professor Dec. 2017 – Nov. 2021
Department of Electrical and Electronic Engineering, Misurata University, Libya

Lecturer Mar. 2013 – Dec. 2017
Department of Electrical and Electronic Engineering, Misurata University, Libya

Assistant Lecturer Sept. 2007 – Mar. 2013

Department of Electrical and Electronic Engineering, Misurata University, Libya

Teaching Assistant Mar. 2003 – Aug. 2007
Department of Electrical and Electronic Engineering, Misurata University, Libya

Adjunct Appointments

- Adjunct Assistant Professor, Libyan Academy for Postgraduate Studies, Mar.–Jul. 2021.
- Adjunct Assistant Professor, College of Industrial Technology, Mar.–Jun. 2019 and Feb.–Apr. 2021.
- Adjunct Lecturer, Civil Aviation College, Misurata, Libya, Sept. 2016 – Jan. 2017.
- Adjunct Lecturer, Higher Institute of Industry, Libya, Sept. 2007 – Jan. 2008 and Sept. 2012 – Jul. 2013.
- Adjunct Lecturer, Higher Institute of Technical Trainers, Libya, Sept. 2007 – Feb. 2008.

Academic Leadership and Administration

Vice Dean for Academic Affairs May 2022 – Aug. 2025
Faculty of Engineering, Misurata University, Libya
Led faculty-level academic affairs: program coordination, academic regulations, quality assurance and improvement, and faculty administration. Honored in Aug. 2025 for leadership achievements in this role.

Head of Department Aug. 2015 – Feb. 2017
Electrical and Electronic Engineering Department, Misurata University, Libya
Directed departmental academic operations, teaching coordination, curriculum development, and administrative planning.

Postgraduate Coordinator Oct. 2017 – present
Electrical and Electronic Engineering Department, Misurata University, Libya
Coordinates postgraduate admissions, supervision arrangements, and program development.

Postgraduate Committee Member Mar. 2019 – Feb. 2022
Faculty of Engineering, Misurata University, Libya

Founder, Department of Renewable Energy 2017
Libyan Academy of Postgraduate Studies, Misurata, Libya
Proposed and established the department and developed the curriculum of its Master's program in collaboration with expert faculty consultants.

Institutional Committees and Service

- Member, Supporting Committee for International University Rankings, Misurata University, 2024.
- Member, Committee for Drafting the Annual Excellence Award Bylaws for Faculty Members and Staff, Misurata University (University President's Decision No. 326 of 2025).
- Member, Faculty of Engineering Syndicate Election Committee, Misurata University.
- Coordinator, Faculty of Engineering Syndicate Election Committee, Misurata University, 2016.
- Member, Preparatory Committee of the Engineering Day, Faculty of Engineering, Misurata University (several years).
- Chair, Electrical and Electronics Engineering Symposium (EEES-2016), Misurata University.
- Member of organizing committees for several local and national engineering conferences in Libya.
- Member, Telecommunication Research Team, Faculty of Engineering, Misurata University.
- Member, Adaptive Antenna Research Team, College of Industrial Technology.
- Member of several academic curriculum development and examination committees.

Publications

Manuscripts Under Review or in Preparation

- **Omar Abu Ella**, "A hybrid multi-agent reinforcement learning framework for robust and scalable RIS control in 6G networks," manuscript in preparation.
- **Omar Abu Ella**, "Performance analysis of reconfigurable intelligent surface systems with imperfect phase shifts," manuscript in preparation.
- Fatma Balto and **Omar Abu Ella**, "Unmanned aerial vehicle base stations: Performance analysis in wireless communications," manuscript in preparation.
- **Omar Abu Ella**, "Performance analysis of double reconfigurable intelligent surface communication systems," submitted to *Transactions on Emerging Telecommunications Technologies*.

Peer-Reviewed Journal Articles

J14. E. Alarbah, S. Joha, M. Abolifa, and **O. Abu Ella**, "Performance evaluation of wireless communication systems assisted by intelligent reflecting surfaces" (in Arabic), *Int. J. Eng. & Inf. Technol.*, vol. 13, no. 2, pp. 27–31, 2025. doi: 10.36602/ijeit.v13i2.539

J13. **O. Abu Ella**, "Performance analysis of the reconfigurable intelligent surfaces communication systems," *J. Commun. Inf. Netw.*, vol. 9, no. 2, pp. 184–191, Jun. 2024. doi: 10.23919/JCIN.2024.10582831

J12. **O. Abu Ella**, "On the capacity of multiple antenna systems," *Annals of Telecommunications*, vol. 79, pp. 437–446, 2024. doi: 10.1007/s12243-023-01000-6

J11. **O. Abu Ella**, "Exact capacity of downlink NOMA system," *Internet Technology Letters*, e342, 2022. doi: 10.1002/itl2.342

J10. **O. Abu Ella** and K. A. Alnajjar, "Mitigation measures for windfarm effects on radar systems," *Int. J. Aerosp. Eng.*, vol. 2022, Art. no. 1083717, 2022. doi: 10.1155/2022/1083717

J9. **O. Abu Ella** and M. Elmusrati, "Achievable rate approximation for massive MIMO with limited number of interfering clients," *Telecommunication Systems*, vol. 79, pp. 463–479, 2022. doi: 10.1007/s11235-021-00871-1

J8. F. Baltu and **O. A. Abu-Ella**, "Introduction to cell-free communication systems with massive MIMO," *Int. J. Eng. Inf. Technol.*, vol. 8, no. 1, pp. 26–32, Oct. 2021.

J7. M. Shabsheb, M. Shabseb, and **O. A. Abu-Ella**, "Challenges and solutions for next generation of millimeter communication" (in Arabic), *Almadar J. Commun., Inf. Technol., Appl.*, vol. 2, no. 1, pp. 8–17, Apr. 2016.

J6. A. Aljerrai, I. Ehtibah, and **O. A. Abu-Ella**, "Interference alignment for multi-user MIMO systems" (in Arabic), *Almadar J. Commun., Inf. Technol., Appl.*, vol. 1, no. 1, pp. 36–44, Apr. 2015.

J5. **O. A. Abu-Ella** and M. Elmusrati, "Recent techniques to cancel and mitigate interference in wireless communication systems" (in Arabic), *Almadar J. Commun., Inf. Technol., Appl.*, vol. 1, no. 1, pp. 9–19, Apr. 2015.

J4. **O. A. Abu-Ella** and M. Elmusrati, "Interference mitigation using optimal successive group decoding for interference channels," *Almadar J. Commun., Inf. Technol., Appl.*, vol. 1, no. 1, pp. 37–54, Jul. 2014.

J3. **O. A. Abu-Ella** and X. Wang, "Large-scale multiple-input–multiple-output transceiver system," *IET Communications*, vol. 7, no. 5, pp. 471–479, Mar. 2013. doi: 10.1049/iet-com.2012.0471

J2. C. Gong, **O. A. Abu-Ella**, X. Wang, and A. Tajer, "Constrained group decoder for interference channels," *Journal of Communications*, vol. 7, no. 5, pp. 382–390, May 2012. doi: 10.4304/jcm.7.5.382-390

J1. **O. A. Abu-Ella** and B. El-Jabu, "Increasing capacity of blind mobile system using pre-filtering technique," *IET Microwaves, Antennas & Propagation*, vol. 2, no. 5, pp. 459–465, Aug. 2008. doi: 10.1049/iet-map:20070174

Book Chapters

B2. **O. A. Abu-Ella** and M. S. Elmusrati, "Recent trends for interference mitigation in multi-antenna wireless systems," in *Handbook of Research on Next Generation Mobile Communication Systems*, A. D. Panagopoulos, Ed. Hershey, PA: IGI Global, 2016, pp. 66–84. doi: 10.4018/978-1-4666-8732-5.ch004

B1. **O. Abu-Ella** and B. El-Jabu, "Adaptive beamforming algorithm using a pre-filtering system," in *Aerospace Technologies Advancements*, T. T. Arif, Ed. InTech, Jan. 2010. doi: 10.5772/7167

Conference Papers

C17. **O. Abu Ella**, "Massive MIMO: In which side should it be to achieve the highest capacity?" in *Proc. IEEE 2nd Int. Maghreb Meeting Conf. Sci. Techn. Autom. Control Comput. Eng. (MI-STA)*, Sabratha, Libya, May 2022.

C16. S. Meftah, I. Almsimit, and **O. A. Abu-Ella**, "Evaluation of the performance of non-orthogonal multiple access (NOMA) technology in the downlink" (in Arabic), in *Proc. 4th Int. Conf. Technical Science*, Tripoli, Libya, 2021.

C15. **O. A. Abu-Ella**, "New design rules to improve helical antenna performance," in *Proc. IEEE Microw. Theory Techn. Wireless Commun. (MTTW)*, Riga, Latvia, 2021, pp. 248–252. doi: 10.1109/MTTW53539.2021.9607134

C14. **O. A. Abu-Ella**, A. Anairia, and M. Zubia, "Pathloss modelling for next generation of millimeter-wave communications," in *Proc. IEEE 1st Int. Maghreb Meeting Conf. Sci. Techn. Autom. Control Comput. Eng. (MI-STA)*, Tripoli, Libya, 2021, pp. 776–781. doi: 10.1109/MI-STA52233.2021.9464521

C13. M. Almagrhy and **O. A. Abu-Ella**, "Low altitude platform communications" (in Arabic), in *Proc. Int. Conf. Technical Science (ICTS)*, Tripoli, Libya, Mar. 2019.

C12. A. Iwhida, M. Kablan, and **O. A. Abu-Ella**, "Massive MIMO modeling for the next generation wireless communication systems," in *Proc. Libyan Int. Conf. Elect. Eng. Technol. (LICEET)*, Tripoli, Libya, Mar. 2018.

C11. A. Anairia, M. Zubia, and **O. A. Abu-Ella**, "Pathloss modeling for 5G millimeter wave communications" (in Arabic), in *Proc. Libyan Int. Conf. Elect. Eng. Technol. (LICEET)*, Tripoli, Libya, Mar. 2018.

C10. **O. A. Abu-Ella** and M. Elmusrati, "Impact of imperfect channel estimation on successive group interference cancellation techniques," in *Proc. 2nd World Symp. Comput. Netw. Inf. Security (WSCNIS)*, Hammamet, Tunisia, Sept. 2015.

C9. **O. A. Abu-Ella** and M. Elmusrati, "Capacity approximation of massive MIMO with optimal successive group decoding system," in *Proc. 8th Int. Conf. Next Gener. Mobile Appl., Services, Technol. (NGMAST)*, Oxford, UK, Sept. 2014, pp. 254–259.

C8. **O. A. Abu-Ella** and M. Elmusrati, "Optimal successive group decoding to mitigate interference in wireless systems," in *Proc. IEEE Int. Conf. Distrib. Comput. Sensor Syst. (DCOSS)*, Marina Del Rey, CA, USA, May 2014, pp. 322–326.

C7. **O. A. Abu-Ella** and M. Elmusrati, "Partial constrained group decoding: A new interference mitigation technique for the next generation networks," in *Proc. 6th Int. Conf. New Technol., Mobility, Security (NTMS)*, Dubai, UAE, Mar. 2014, pp. 1–5.

C6. **O. A. Abu-Ella** and X. Wang, "Interference mitigation via constrained partial group decoding for uplink multicell MIMO systems," in *Proc. Int. Conf. Electron. Commun. Eng. (ICECE)*, Istanbul, Türkiye, Jun. 2013, pp. 1810–1813.

C5. **O. A. Abu-Ella** and B. El-Jabu, "Optimal robust adaptive LMS algorithm without adaptation step-size," in *Proc. Global Symp. Millimeter Waves (GSMM)*, Apr. 2008, pp. 249–251.

C4. B. A. El-Jabu, J. A. Srar, and **O. A. Abu-Ella**, "Beamforming in tight specifications environment using generalized minimum mean error (GMME) algorithm," in *Proc. IEEE Aerospace Conf.*, Mar. 2007, pp. 1–7.

C3. J. A. Srar, **O. A. Abu-Ella**, and B. A. El-Jabu, "Beamforming in tight specifications environment," in *Proc. IEEE Aerospace Conf.*, Mar. 2007, pp. 1–6.

C2. **O. A. Abu-Ella** and B. A. El-Jabu, "Performance improvement of blind adaptive beamforming algorithms using pre-filtering technique," in *Proc. IEEE Aerospace Conf.*, Mar. 2007, pp. 1–4.

C1. J. Eljaroshi and **O. A. Abu-Ella**, "The strategic importance of technological cities to developing countries and their role in supporting and settling advanced technologies" (in Arabic), in *Proc. Technical Cities Symp.*, National Office of Research and Development, Tripoli, Libya, Oct. 2003.

Graduate Supervision

Master's Theses Supervised or Co-Supervised

- Next-generation communication systems with unmanned aerial vehicles (UAVs).
- Deep learning-aided reconfigurable intelligent surfaces (RIS).
- Service-oriented network slicing for next-generation mobile networks (NGMN).
- Cell-free massive MIMO for next-generation wireless communications.
- Reconfigurable intelligent surfaces for next-generation wireless communications.
- Low Earth orbit (LEO) satellites in next-generation communications.
- Study and development of long-range radars: The P18 as a case study.

Teaching Portfolio

Postgraduate Courses

Next Generation Wireless Systems; Multiple Antenna Systems; Numerical Analysis.

Undergraduate Courses

Signals and Systems
Communication Theory I
Optical Fiber Communications
Antenna Theory
Communication Systems Simulation and Modeling Lab
Physics II
MATLAB Basics
Communication Systems
Wireless Communications
Electromagnetic Fields I
Electrical Measurements and Instrumentation
Computer Application and Design Laboratory
Numerical Analysis

Teaching Assistant and Training Experience

- Assisted in Mathematics I, Numerical Analysis using MATLAB, Linear Systems, Electronics II, Automatic Control, and the Simulation of Communication Systems Laboratory.
- Instructor of intensive MATLAB training courses for faculty members, graduate students, and undergraduates at several institutions.

Editorial and Professional Service

- **Editorial Board Member:** *Almadar Journal for Communications, Information Technology, and Applications* (AJCITA, ISSN 2313-156X).
- **Technical Program Committee Member:** IEEE ISWTA 2012 and ISWTA 2014.
- **Journal Reviewer:** IEEE Transactions on Wireless Communications; IEEE Transactions on Vehicular Technology; IEEE Journal on Selected Areas in Communications; IEEE Communications Letters; IEEE Wireless Communications Letters; IEEE Antennas and Wireless Propagation Letters; IET Communications; IET Networks; IET Electronics Letters; Wireless Networks (Springer); International Journal of Physical Sciences.
- **Conference Reviewer:** IEEE ISWCS, IEEE VTC, IEEE ICC, IEEE ICASSP, IEEE SPAWC, IEEE MILCOM, IEEE WCNC, IEEE JSTSP, ICSPCS, ISWPC, MIC-CCA, TCOM-TPS, and ISWTA.

Awards and Honors

- Honored for leadership achievements as Vice Dean for Academic Affairs, Faculty of Engineering, Misurata University, Aug. 2025.
- Libyan Ministry of Higher Education Scholarship, 2007.
- Recognized for contributions to establishing the Faculty of Engineering, Misurata University, 2005.
- Top of class, academic years 1999/2000 and 2000/2001.

Professional Affiliations

- IEEE Member since 2006 (more than ten years of membership).
- Member, American Association for Science and Technology (AASCIT), 2015.
- Former Member, IET, 2007.

Selected Professional Training

- *ENBRAIN: Building Capacity in Renewable and Sustainable Energy for Libya*, co-funded by the Erasmus+ Programme of the European Union, coordinated by Politecnico di Torino, Oct. 2019.
- *Electrical Engineering Principles and Power Systems Equipment*, TQ Education and Training Ltd., Nottingham, United Kingdom, Mar. 2003.

Languages

Arabic (native); English (fluent, professional and academic working language).

References

Available upon request.